Here we represent figures of different sections of VIRMOTIF:

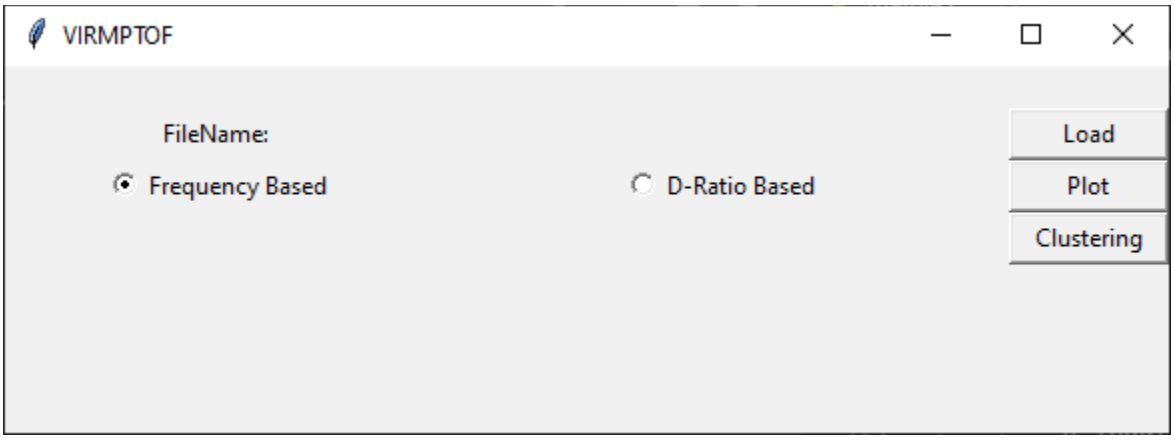


Figure 1. VIRMOTIF main page for loading datasets and choosing between Plot and Clustering

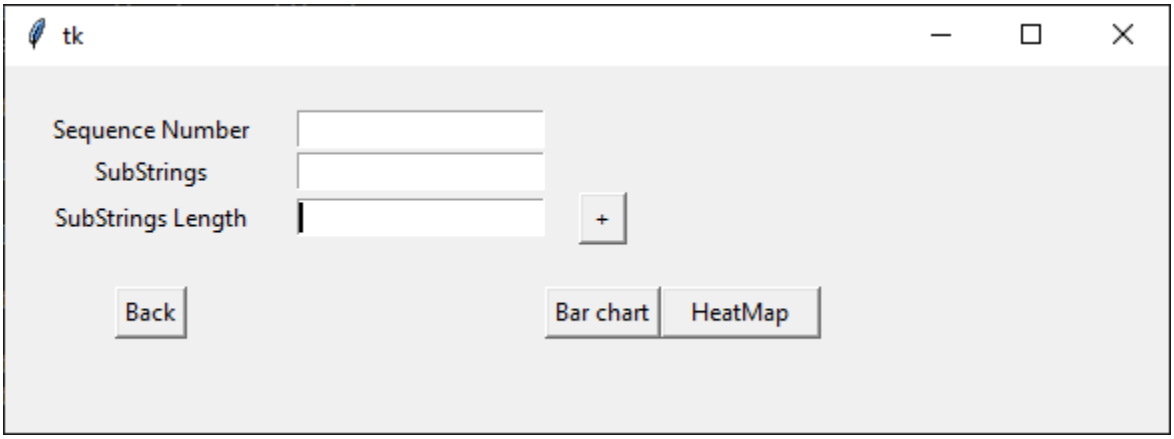


Figure 2. Plot page containing bar chart and heatmap for Ploting

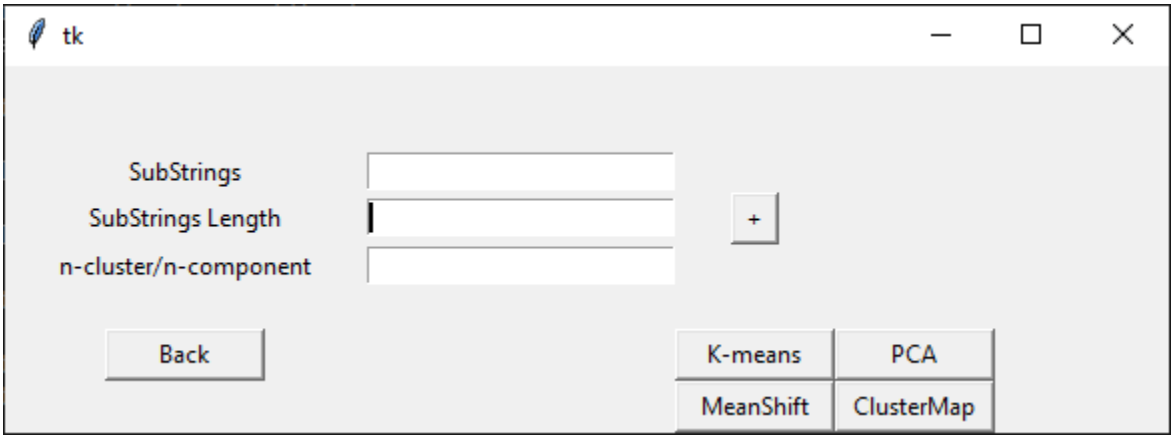


Figure 3. Cluster page containing Kmeans, Meanshift and ClusterMap for clustering , and PCA as a reduction feature

Here we represent several examples to provide more information on how VIRMOTIF works.

- **Bar Chart**

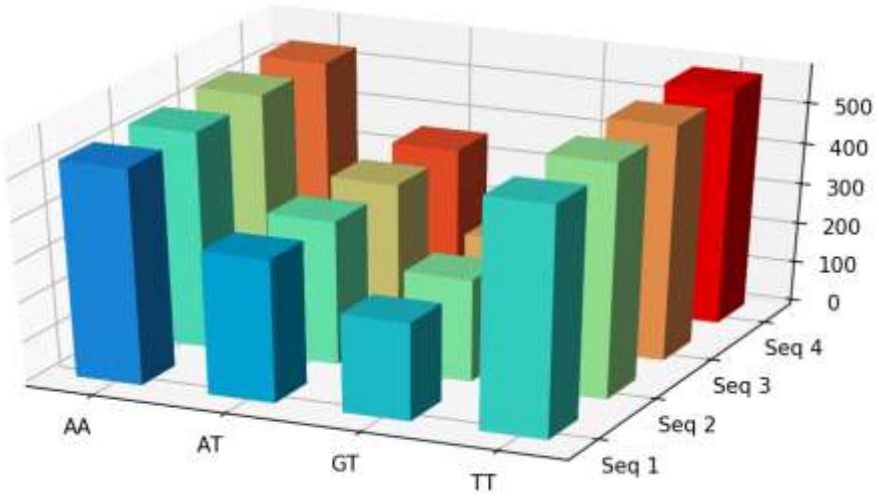


Figure 4. Visualization of first 4 sequences with substring AA, AT, GT, and TT. In this example we use frequencies as computation method.

- **K-Means**

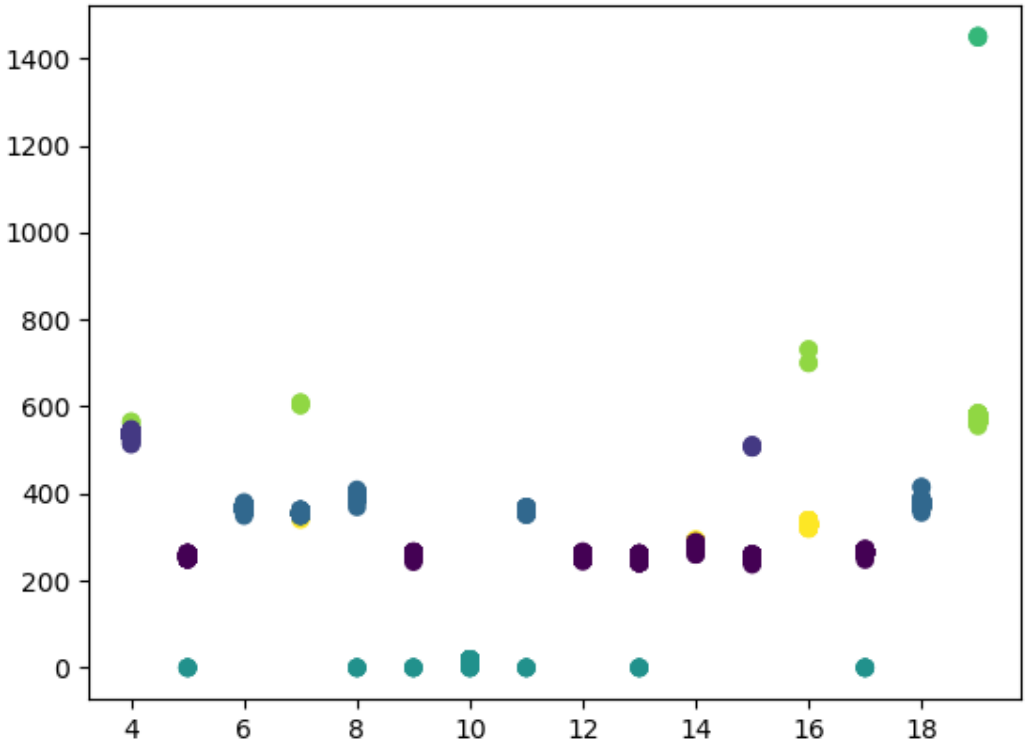


Figure 5. For this case, we select frequency for computing and substrings with length 2. Also, we chose to cluster with Kmeans and number of clusters to be 7.

- **MeanShift**

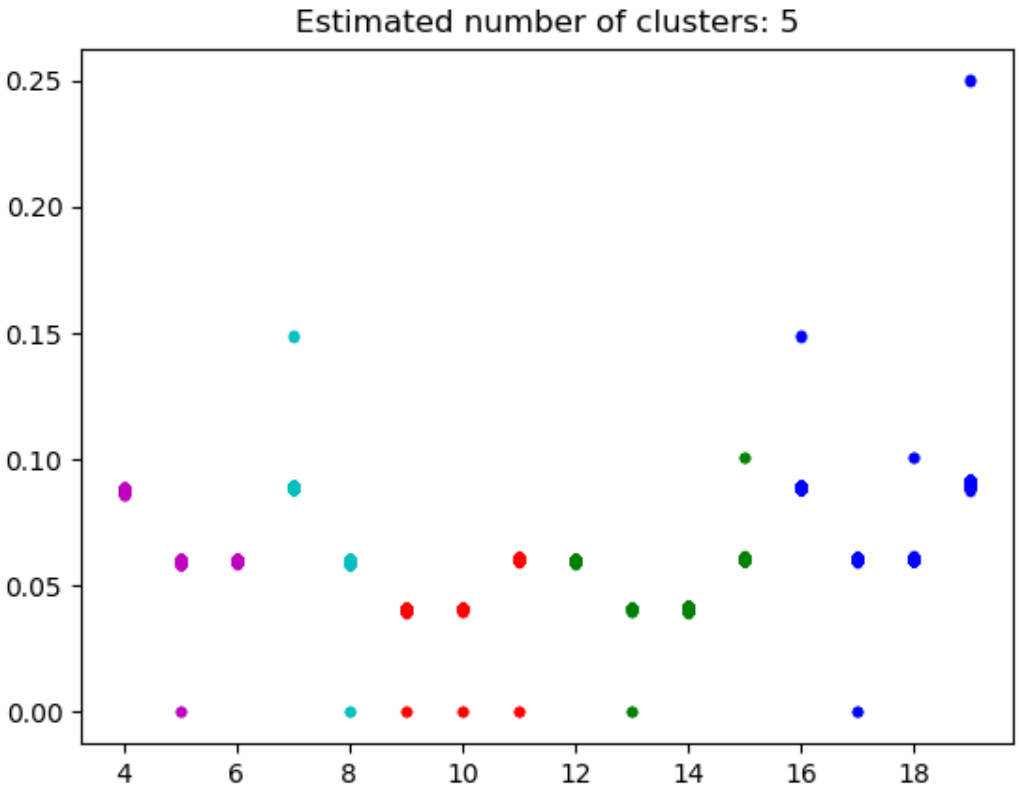


Figure 6. Here we selected D-ratio for computing and chooses substrings with length 2. We also used Meanshift to identify the optimal number of clusters. For this case, Meanshift identifies 5 as the number of optimal clusters.

- **Cluster Map**

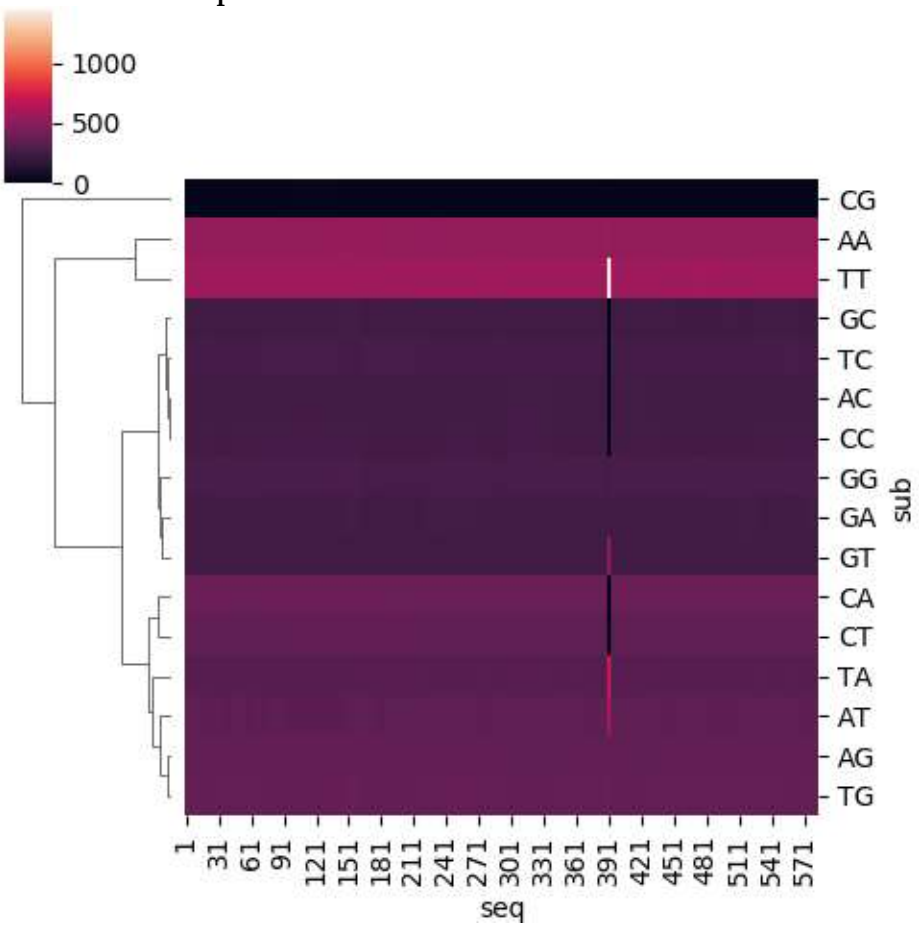


Figure 7. Here we select D-ratio for computing the motifs and substrings with length of 2 and heatmap cluster method to cluster the data.